How can Artificial Intelligence aid Critical Thinking skills in First-Year UK University Students with Limited Academic Writing Experience?

Course: MA Digital Teaching and Learning

Module: EDUC4062 Social Contexts of Learning

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Word count: 3100 words

Date: 30/04/2026

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Needs Analysis

Current Artificial Intelligence Trends in Higher Education

Artificial intelligence (AI) usage has surged globally, with ChatGPT reaching 1 million users in 5 days and 100 million in two months (Lendvai, 2025). Even within Higher Education (HE), most students now use AI in their assessments (Freeman, 2025).

AW is a particularly prevalent HE assessment method, supporting analytical reasoning, argumentation and disciplinary knowledge (Gabay et al., 2026). AI used as a research partner has potential to improve HE students' AW (academic writing), language and creativity (Bennett & Abusalem, 2024), but students adopting it for summative assignments without scrutiny can challenge academic integrity, ethics and the principles of co-authorship.

This paper therefore seeks to respond to the growing gap in HE caused by students' swift AI adoption without solid theoretical or pedagogical foundations; the challenge lies in how to structure AI usage to preserve and improve students' CT in HE.

Significance of Critical Thinking (as a Skill and in Academic Writing)

Critical Thinking (CT) is one possible intervention to the human mind's default inclinations towards unreliable prejudice, egocentricity, bias and irrational thinking (Elder, 2022). Without intentionally continuous CT training, individuals are less likely to intervene when stuck in sociocentric patterns of thought that contort information to better fit their comfort zones; if these natural tendencies persist, they ultimately prevent learners from developing the higher-order thinking skills (HOTS) that can contribute meaningfully to society.

Within the HE context, the HOTS of CT can be defined as "the intellectually disciplined process of actively and skilfully conceptualising, applying, analysing, synthesising, and evaluating information gathered from observation, experience, reflection, reasoning, or communication, as a guide to belief and action" (Scriven & Paul, 1987, as cited in Elder, 2022, p.8). This definition of CT demands intellectual standards around clarity, accuracy and logical coherence, which overlaps with AW good practices (Elder, 2022). It also aligns with common AW processes in which students are expected to engage; identifying a purpose, evaluating sources, constructing arguments, and drawing

reasoned conclusions. Consequently, AW is not merely about delivering ideas, but a practice through which CT is developed and demonstrated.

Despite this close relationship between AW in HE and CT skills globally, the latter is rarely implemented within an effective pedagogical framework (Andreucci et al., 2023). Van der Zanden et al. (2020) found that many educators hold varying understandings of CT, thus fostering it inconsistently in their teaching. Consequently, many students still leave academic programmes without developing the reasoning skills required for HOTS (Van Damme & Zahner, 2022).

Developing CT requires implicit reasoning processes to be made explicit, to avoid individuals engaging in domain-specific thinking without understanding or articulating its underlying cognitive processes (Elder, 2022). In HE, AW can serve as a key medium for externalising thinking; if learners make their reasoning visible, structured, and open to reflection through this process, they not only develop writing skills but also gain a clearer understanding of how to construct, evaluate, and regulate their own thinking.

Impact of Artificial Intelligence usage on Critical thinking in Academic writing

AI can now support students through every AW stage, from correcting grammar to improving argumental coherence through structure (Gerlich, 2025). It can also serve as a research assistant, identifying credible sources and then summarising, synthesising, and analysing complex academic literature to pinpoint research gaps or suggest potential counterarguments. AI can even generate entire academic papers now with minimal prompting and human intervention (Lendvai, 2025).

These extensive levels of AI support may, however, cause less autonomous, motivated and self-regulated students to over-rely on it for AW, blurring the line between ownership of authorship (Cui et al., 2025). In fact, differences in learners' self-regulated learning have been shown to play a critical role in AI-supported writing, exerting a stronger influence on performance than AI literacy (Shi et al., 2025). This suggests that current AI usage patterns may undermine the fundamental role of CT to provide learners with standards for regulating their own thinking processes. Students' current AI usage therefore raises concern for overall CT development, especially within the realms of metacognition, productive struggle (PS), cognitive debt (CD) and scaffolding.

AI that provides immediate and well-formed outputs reduces the need for learners to articulate and examine their own reasoning, thereby limiting opportunities for metacognitive reflection and self-regulation. Zimmerman's (1990) work on self-regulated learning posits that students need to apply metacognitive strategies such as reflective thinking, planning and monitoring to regulate their learning. For AW learners, this should include appropriately planning, monitoring and evaluating their writing strategies (Hammann, 2005). This usually does not happen unprompted with AI; as learners increasingly offload cognitive effort to AI systems, engagement in deep and reflective thinking declines (Gerlich, 2025).

These AI-bypassed, effortful processes are referred to by Hiebert & Grouws (2007) as 'productive struggle' (PS), required by students to make sense of complex problems and construct deeper conceptual understanding. When learners engage in the necessary cognitive effort of PS, they build the internal structures necessary for independent reasoning (Hiebert & Grouws, 2007) and therefore CT. Learners bypassing the PS process may perform well in the short term whilst still failing to develop these underlying structures. Over time, this results in the accumulation of cognitive debt

(CD), “a condition in which repeated reliance on external systems [like AI] replaces the effortful cognitive processes required for independent thinking…” resulting in “...diminished critical inquiry, increased vulnerability to manipulation, decreased creativity." (Kosmyna et al., 2025, p. 151).

Such AI support may also reduce the benefits found in traditional teaching, such as the expert thinking insights and support that scaffolding offers during writing skill development, with assistance gradually withdrawn as learners gain competence (Wood et al., 1976). This is known as ‘fading’, which is conspicuously absent in many of the presently popular AI models.

As a result of this current AI usage that bypasses metacognitive PS and scaffolded CD, AW is vulnerable to this risk of diminished CT skills. When AI removes the 'desirable difficulties' (Bjork and Bjork, 2011) of students articulating and structuring an argument, it disrupts the process of externalising thinking that PS makes possible. Consequently, learners are deprived of the struggle needed to develop the internal evaluative standards (Elder, 2022) required to critically oversee AI-generated output. This paper therefore argues that, without intervention, current AI usage trends are likely to deepen dependency, rather than cultivate the independent reasoning that HE currently demands.

Learning needs

Based on the above theoretical analysis of cognitive offloading and the necessity of PS, three primary learning needs have been identified:

1. **Promote Effortful Cognition:** To counteract the risks of immediate AI outputs, learners need environments that encourage effortful cognitive processes. This involves preserving the PS necessary to internalise academic standards and reduce CD, rather than passively relying on AI-generated results.
2. **Enhance Metacognitive Regulation:** There is a critical need to strengthen metacognitive awareness and self-regulation. Since self-regulatory skills influence performance more than AI literacy, learners require instructional scaffolding, such as reflective prompts and nudges, to critically oversee their interaction with AI.
3. **Scaffolding to Externalise the Writing Process:** Learners must develop academic writing skills by externalising their thinking. This involves moving beyond surface-level performance

to understand the structural and logical essence of academic discourse, using AW as a visible medium to construct, evaluate, and refine their own reasoning.

Design

Design Framework

Project *Thamus* is an AI-assisted desktop application, designed to promote CT (Critical Thinking) during the academic reading (AR) and writing (AW) process by engaging students in metacognitive reflection and scaffolded PS, instead of offloading their cognitive processes to AI (Griffith & James, 2025; Hiebert & Grouws, 2007).

Design Goals

Three core design goals were identified to support *Thamus'* AI-assisted AW process for learners:

1. To preserve PS, Thamus is designed to act as a guide and provide responses as questions, checklists and reflective prompts, instead of generating immediate output. This ensures that cognitive work remains with the student.

2. As previously discussed, current AW requires metacognitive strategies such as reflective thinking, planning and monitoring (Zimmerman, 1990) to help regulate learners' writing by appropriately planning, monitoring and evaluating their writing strategies (Hammann, 2005). This may not occur automatically in most present AI models and therefore needs appropriate conditions, such as use of nudges, to support self-regulated learning (Kravčík & Klamma, 2014). Consequently, *Thamus* nudges the learners towards goal setting and self-reflection through prompts, that encourage them to pause, monitor, evaluate, and appropriately develop their writing process.

3. Wood, Bruner and Ross's (1976) concept of instructional scaffolding and Vygotsky's (1978) notion of the 'More Knowledgeable Other' (MKO) underscore the role of tutors in facilitating an individual's learning process. In the AW process, tutors may support learners by making writing and its strategies visible, guiding them through the procedural steps and gradually removing support as learners develop those skills (Englert et al., 2006). Thus, *Thamus* is

designed to act as the KO that externalises the writing process across multiple stages, guiding learners through writing activities, outlining customised tasks and offering writing feedback. Scaffolding is gradually faded as Thamus moves from guiding the writing process to nudging the learner to attempt writing tasks independently. This scaffolded Zone of Proximal Development (Vygotsky, 1978) encourages students to push through their cognitive incongruity to develop HOTS, such as reasoning skills (Hatano, 1988, as cited in Hiebert & Grouws, 2007) needed for CT.

Thamus' Stages

Successful AR and AW require critical reading (CR) and critical writing (CW) skills respectively, the dimensions of which are intertwined (Wallace & Wray, 2021). The entire process has therefore been mapped across five phases adapted from Bloom's Taxonomy; Purpose, Exploration, Investigation, Synthesis, and Proofreading) (Anderson et al., 2001). As Bloom's Taxonomy organises cognitive complexity hierarchically, it aligns with the design goal of gradually increasing PS.

1. ***Context Setting and Purpose***

This stage establishes the foundation for the entire writing process by helping students develop a clear understanding of their writing task, contextualised by their personal goals and challenges. It comprises of two interrelated activities:
(A) Understanding the Writing Task and ***(B)*** **Motivation and Challenges.**

(A) Understanding the Writing Task is grounded in cognitive process theories, that effective writers should put effort in problem representation **before** drafting (Baaijen & Galbraith, 2018). Rephrasing the assignment in their own words helps students actively process the task, identify comprehension gaps, and clarify its scope (Wingate, 2012). Students upload their rubric and instructions to *Thamus*, which analyses the materials, generates exploratory questions to surface misunderstandings, and produces a task summary for learner confirmation (see Appendix 1.1).

(B) Motivation and Challenges function as a metacognitive starting point, grounded in self-regulated learning theory: identifying personal goals and apprehension **before** writing activates

motivational resources and helps direct learners' attention strategically (Mega et al., 2014; Panadero, 2017). Niemiec & Ryan (2009) further indicate that conversational, low-stakes reflection on anxiety can support intrinsic motivation and encourage students to clarify their thinking, setting motivation, relevance, and direction before any content mechanics appear (Wallace & Wray, 2021). *Thamus then* generates targeted prompts (see Appendix 1.2), storing students' responses in a learner profile to personalise feedback across **all** subsequent stages, closing the metacognitive loop by rechecking goal achievement and areas of concern in the final proofreading stage 5 (see Appendix 5).

2. Exploration

This stage aims to coax out learners' existing topic knowledge, including assumptions and gaps in understanding, then aid them in drafting a research strategy for the most relevant resources. This stage entails two activities: ***(A) Freewriting*** and ***(B) Guided Research Strategy*****.**

(A) Freewriting refers to writing continuously about a specific topic and is considered a powerful technique to develop writing skills (Elbow, 1998). Spontaneous writing can help ease the burden of simultaneously thinking about the topic and how to best articulate it (Li, 2007). In this activity, students' metacognitive skills are developed by inviting them to reflect on their topic before engaging with academic resources. This allows them to discover ideas, identify gaps and questions to guide their research. *Thamus'* freewriting activity is scaffolded by first describing it, followed by activity instructions customised to learners' contexts, nudging them to submit their response. The instructions entail a list of prompts to support the process, with guiding constraints such as recommended word and time limits. After submitting their freewriting response, Thamus gives feedback indicating whether learners have submitted a satisfactory response. If it does not meet the freewriting criteria, then they are asked to resubmit their response

(See Appendix 2.1).

(B) Guided Research Strategy focuses on helping students develop a research strategy. This refers to academic and information literacy development, whereby students learn how to search and select quality resources for AW (Castillo-Martínez & Ramírez-

Montoya, 2021). In this activity, Thamus outlines different skills that students need to consider before initiating research, such as identifying credible databases and resources and keyword search techniques, inviting students to select which skills they wish to learn. Based on their response, *Thamus* describes the skill and outlines a customised task to guide learners' independent research. Here, they may be asked to outline the databases they are considering, search filters they plan on implementing and parameters for selecting readings. If students' responses appropriately address the task, then *Thamus* may direct them to the next skill or activity. If, however, students' responses indicate misconception, *Thamus* offers feedback and encourages the student to resubmit their response (See Appendix 2.2).

3. Investigation

After students have checked their research strategy with *Thamus* they need to implement it, relating those newly found readings to their assignment's question(s), scope and thematic topics. They will upload all their selected readings or related notes before answering *Thamus'* ***(A) Investigative Questions*****,** then ***(B) Connective Questions*** *about the papers*.

(A) Investigative Questions from *Thamus* will initially guide students to critically evaluate the **general** relevance of their chosen papers *or* related notes. This metacognitive process allows students to monitor and reflect on their progress when justifying their choice of readings to *Thamus* (see Appendix 3.1). If students struggle to answer, these questions can also **nudge** them on CR of **individual** papers; *Thamus* can direct students to the papers' abstracts to summarise key points but it will not extract those **for** the students, to avoid possible AI hallucinations and still encourage PS. Students will be given a space separate to the chat box to record their answers, such as an editable table or mind-map, to which they can refer back to later.

(B) Connective questions from *Thamus* will be based on students' previous investigative answers, connecting their papers to **one another** (see Appendix 3.2). To help students better visualise those overall connections, they can edit their previous notes while answering those questions. For instance, they may draw mind-map arrows between related

arguments from different papers, to connect previously scattered insights into thematic understanding.

4. Synthesis

Here, students bring together the thematic understanding from their previous subject ideas and researched arguments, into a logical structure befitting of AW requirements. Consequently, this stage focuses on HOTS for overall AW ***coherence***, rather than lower-order, cohesive AW skills. Academic coherence refers to the **macro-level** organisation of an argument, such as its overall stance and paragraph topics (Smith, 2025).

(A) The Essay Outline should first be populated with students' own research notes, such as those created in the previous stage (see Appendix 4.1). To encourage PS, *Thamus* will first see how much of the table students are able to fill out by **themselves**.

(B) Argumental Questions can nudge any students who struggled to populate the outline appropriately, helping them hone their thematic insights specifically to an **AW context** as they write up their initial paragraphs (see Appendix 4.2). Students' Stage 1 assignment prompts and Stage 3 notes will still be visible and expandable, to keep their draft focused.

(C) A reverse outline will be co-created based on students' above "inquiry-based writing instruction" (Kim et al., 2025, p. 1279) and *Thamus'* follow-up prompts, from which students can clarify their overall essay **flow** (see Appendix 4.3). They may use this reverse outline to amend, clarify or add to their initial outline as appropriate, such as adjusting their paragraph topics to better align with their overall argument direction.

(D) PEEL builder prompts will be populated according to *Thamus'* understanding of each paragraph's subtopics, based on the students' overall paragraph delineation and topic sentences (see Appendix 4.4). By **externalising** this visually for validation, *Thamus* gives students the chance to correct any missing or unclear information, metacognitively monitoring their own writing from initial argumentation to a developing AW format. This macro-level outlining, paragraphing and PEEL process should stand students in good stead to next tackle the micro-level cohesion required in the last stage.

5. Proofreading

This is the culminating phase in which students receive targeted feedback on their completed draft. It is an important part of the AW process that includes lower-order skills, such as overall proofreading, formatting, referencing and citing. Students will be able to customise these according to their field's norm (e.g., APA 7th, Chicago) when submitting their draft.

Thamus is deliberately structured so that learner self-assessment **precedes** AI review. Using a sliding Likert scale, students rate their confidence across key dimensions before receiving feedback, building the evaluative habits that underpin long-term writing development and academic self-efficacy (Panadero et al., 2018). This sequencing reflects Yan & Brown's (2017) established finding that self-evaluation **before** external feedback strengthens self-efficacy and develops critical awareness, aligning with the view that self-monitoring is a core metacognitive skill which improves learning outcomes (Dignath & Büttner, 2008). Rather than correcting errors directly, *Thamus* highlights problematic areas and nudges students to identify and articulate the issue themselves, personalising suggestions based on their Stage 1 learner profile (see Appendix 5).

Conclusion

This proposed design forms the first part of an iterative, design-based research process. Its first phase examined the impact of HE students' current AI usage on CT skills, revealing a pedagogical gap between AW aims and CW skills. Students offloading key CW aspects to AI limits their authorship and meaningful learning by reducing the PS essential for CT development within AW.

The proposed AI intervention, *Thamus*, is thus designed to support learners in reclaiming their autonomy over the AW process through adaptive scaffolding. *Thamus*' dynamic writing plans, checklists and dialogic questioning are personalised to students' goals and anticipated challenges, acting as a cognitive mentor and MKO that sustains PS instead of replacing the cognitive processes essential for deeper learning in HE.

The objective of the next phase is to obtain initial design feedback from the intended user groups, students and teaching staff. Their evaluation of *Thamus'* anticipated usage, and its capacity to address the previously identified gaps in AW pedagogy, will inform *Thamus*' core design specifications. Subsequently, the first prototype for user testing shall be built based on these target user group insights.

Appendix:

Glossary (of Acronyms)

- AI (artificial intelligence)
- AR (academic reading)
- AW (academic writing)
- CD (cognitive debt)
- CR (critical reading)
- CT (critical thinking)
- CW (critical writing)
- GenAI (generative AI)
- HE (Higher Education)
- HOTS (higher-order thinking skills, such as CT)
- LLM (large language models; see AI)
- MKO (More Knowledgeable Other; see Vygotsky, 1978)
- PS (productive struggle)

Sample Activities (from *Thamus'* 5 Design Stages)

1. Appendix 1 - STAGE 1 – Context-setting and Purpose

***Appendix 1.1* – Understanding the Writing Task**

[Students upload their rubric and assignment instructions to *Thamus*, which analyses the materials and asks exploratory questions to develop a shared understanding of the task (see sample below). Throughout the dialogue, students articulate the task in their own words, surfacing any comprehension gaps in comprehension and confirming a task summary that is stored in their learner profile for all subsequent stages.]

Sample prompts:

- Please upload your rubric and assignment instructions so I can understand what is being asked of you.
- What type of writing is this, and what is the central question you need to address?
- What do you think your teacher is looking for most based on the rubric?
- Does this seem right to you? Is there anything you want to add?

***Appendix 1.2* – Motivation and Challenge Task**

[Students identify their writing goals and areas of difficulty through a short reflective dialogue with *Thamus* (see example below). Responses are recorded in the learner profile to personalise feedback across all subsequent stages, and then revisited in Stage 5 to close the metacognitive loop.]

Example questions:

- What is your goal for this writing task? You can select from the options below or describe it in your own words: [Improve my writing skills/ Achieve academic excellence/ Explore the topic in depth/ All the above].
- Good! Now, from the areas below, which are you most worried about? [Structure and Augmentation/ Academic language and tone/ Critical Analysis/ Referencing and citations/ Other].
- That's really helpful to know; you are not alone in finding [Critical Analysis] tricky. So, your goal is [academic excellence], and your main areas of concern are [critical analysis and APA referencing]. Does that seem right? Is there anything else you wish to add

2. *Appendix 2 - STAGE 2 – (Subject) Exploration*

Appendix 2.1 – Freewriting task

Now that we have understood your writing task, the first step is to look within! Freewriting is an unstructured writing technique, where you write continuously about a topic without worrying if your writing is 'correct'. The goal is to discover what you know and what you are uncertain about.

Task Guideline: Your task is to write freely for 20 minutes and produce minimum of 200 words. You may use the following prompts to guide your thoughts:

- What do you already know about this topic?
- What would you like to know more about the topic?
- What would someone who disagrees with this topic argue with you?
- What assumptions might you be making about this topic?
- What evidence do you have to support your view?
- Which readings from your course feel relevant to this topic?

Appendix 2.2 – Research strategy Task

Now that we have explored your initial thoughts on your topic, it is time to think about how you will find academic sources to support your argument.

What is a research strategy? A research strategy is a deliberate plan for how you will search for and select credible academic sources. In academic writing, this involves developing your information literacy (that is, knowing where to look, how to search effectively, and how to judge whether a source is appropriate for your purpose). Having a clear strategy before you begin saves time, by ensuring the evidence you gather is relevant and trustworthy.

Before you begin your research, which of the following skills you would like to learn about? Select all that apply:

- **Keyword search techniques:** How to construct effective search terms to find relevant literature
- **Identifying credible databases:** How to select appropriate academic databases for your topic

- **Evaluating and selecting sources:** How to assess whether a source is reliable and relevant to your essay

[If student selects: 'Identifying credible databases]

Not all databases carry the same academic weight. Knowing which to consult and how to distinguish between source types is a foundational research skill. Academic databases host peer-reviewed literature, meaning the work has been evaluated by experts before publication; this makes them the most reliable starting point for any academic essay.

Types of sources and where to find them [customised to learners' contexts]:

Source Type	Where to Find It	Best Used For
Peer-reviewed journal articles	JSTOR, Scopus, Web of Science, PubMed	Theoretical frameworks, empirical data, scientific evidence
Academic books and book chapters	Google Scholar, your university's library catalogue	In-depth, conceptual and political discussions
Grey literature	Government websites, UN, IPCC, NGO publications	Policy context, real-world data, applied evidence
Institutional and think tank reports	Chatham House, Brookings Institution, World Bank	Economic and developmental perspectives on climate policy
News and journalism	The Guardian, BBC, The Economist	Background context and current events (tip: use sparingly)
General web sources	Open websites, blogs, social media	Background context only (not appropriate as an academically cited source)

Task Guidelines: Now it is your turn. Using what you have just learned, respond to the following:

Set aside 30–45 minutes for this activity. Identify **at least three databases** you are planning to use and briefly explain your reasons for selecting each one. Alongside this, list **at least three**

parameters you will apply to evaluate whether a source is suitable for your essay, as well as **five keywords** you are planning to use in your search. Remember, the decisions about which sources to consult are **yours** to make; *Thamus* is here to guide your strategy, not to make those choices for you.

[If student response appropriately addresses the task]

***Thamus*:** Well done! Your response demonstrates a clear understanding of how to approach your research. Let us now move on to the next skill or activity.

[If student response indicates misconceptions]

***Thamus*:** Thank you for your response. Before we move on, there are a couple of things worth revisiting. [Targeted feedback based on misconception(s)]. Take a moment to revise your response and resubmit when you are ready; there are no penalties for doing so, and refining your thinking at this stage will strengthen your research.

3. *Appendix 3 - STAGE 3 – Investigation (Implementing the Research Strategy)*

***Appendix 3.1* – Sample Investigative questions:**

Now we need to decide **how** the papers you've uploaded support, challenge or add nuance to your assignment question(s)/ prompt(s). Please use the note-taking space below to answer the following:

- Which idea(s) from the readings **challenge** your assignment questions/ prompts? How do they do this (e.g. do they have peer reviewed sources, as previously discussed)?"
- Which arguments from the papers were the **least** convincing to you and why?
    - [Nudge, if students are still struggling to critically evaluate]: Are there any limitations that could affect the validity of this paper's claims in relation to your assignment questions/ prompts? (e.g. how generalisable are these results to your X population in the question?)

Appendix 3.2, Sample Connective questions:

Now we've decided how your uploaded papers support, challenge or add nuance to your assignment question(s)/ prompt(s), we need to start connecting similarities and differences **across** the papers. This will help to later build your argument:

- Based on your previous answers, can you please group all the ideas/ arguments/ evidence which **support** your assignment questions/ prompts? Don't forget to cite the relevant papers (You can edit your answers directly in the pop-up window)
    - Within all these **supporting** arguments/ ideas/ evidence, do you see any **sub-categories** emerging? (Feel free to drag and drop them within the pop-up table to categorise accordingly)
- Rereading the notes you have taken so far?
    - Do any papers counterargue one another?
        - If so, give a 1-sentence summary of their thematic relation.

4. Appendix 4 - STAGE 4 – Synthesising Arguments

Appendix 4.1 – Sample essay outline:

Please fill this out by **cross-referencing** the arguments, ideas and evidence from your previous [research table/ mindmap/ notes]:

<table>
<tr><td colspan="3">Title:
(Tip: come back to this at the end, once you know your overall stance)</td></tr>
<tr><td colspan="3">Thesis Statement:
(Tip: come back to this once you know your overall arguments and counterarguments)</td></tr>
<tr><td colspan="3">Introduction:
(Any important background your audience needs to know first, to help them make sense of the below arguments?)</td></tr>
<tr><td>Argument A:</td><td>Argument B:</td><td>Argument C:</td></tr>
<tr><td>For or against?</td><td>For or against?</td><td>For or against?</td></tr>
<tr><td>Summary:</td><td>Summary:</td><td>Summary:</td></tr>
<tr><td>Source:</td><td>Source:</td><td>Source:</td></tr>
<tr><td>Factors to consider (e.g. limitations):</td><td>Factors to consider (e.g. limitations):</td><td>Factors to consider (e.g. limitations):</td></tr>
<tr><td>Counterargument:</td><td>Counterargument:</td><td>Counterargument:</td></tr>
</table>

Source:	Source:	Source:
Factors to consider (*e.g. limitations*):	Factors to consider (*e.g. limitations*):	Factors to consider (*e.g. limitation*s):
Any Other Points: (e.g. overall limitations, in relation to the assignment context?)		
Conclusion: (Tip: come back to this once you know your overall arguments and counterarguments)		

[Possible questions to be asked if students struggle with filling in the above essay outline]:

Appendix 4.2 **– Sample Argumental questions:**

- Based on your notes, which order of [arguments A/B/C] aligns with you most? Feel free to [number/ add/ rank] them accordingly.
- How does [argument A] connect to [Argument B]? Is it a counterargument (for the opposite side) or a separate point (for the same side)?
- Based on all these arguments, would you say you're more **for** or **against** [assignment question/ prompt] overall? (Tip: That's your stance!)
- If someone asked you to summarise this essay in one sentence, what would you say? (Tip: That's your thesis statement!)

Appendix 4.3 **– Sample Reverse Outline prompts:**

Here is everything that you have written thus far. Now:

- Add headings to summarise what each section is about
- [*Thamus* hides everything except for the headings] If you read only these headings, what would you think this paper was about? What would you assume was the overall stance – positive or negative?
- Would you like to change any of the headings (or their order) based on this?"
- Can you number these headings according to the order they would appear in your paper? (Tip: Could you use this order for your paragraphs?)

***Appendix 4.4* – Sample PEEL builder prompts**:

Based on the paragraphs you finalised above (from the reverse outline), here is what I understood each one to be about:

- Point:
    - Topic sentence – Summary of this paragraph's argument?
- Example:
    - Evidence – source (cited)?
- Explanation:
    - So what? How does this point relate back to the assignment question/ prompt?
- Link:
    - How does this directly answer the assignment question/ prompt, or introduce any counterarguments (in next paragraph)?

5. Appendix 5 - STAGE 5 – Proofreading Task

[Before receiving any AI feedback, students first self-assess their draft using a sliding Likert scale. They then upload their draft and select a priority area for review. *Thamus* provides annotated, personalised feedback through exploratory nudges
(see examples below) rather than direct corrections, then compares its review with the student's self-assessment to generate a targeted revision plan, closing the metacognitive loop opened in Stage 1.]

<u>Example questions:</u>

- Before I review your draft, rate how confident you feel about each of these areas: [criticality/ coherence/ academic tone/ referencing/ formatting…. (Sliding scale can be added as 2/5, 3/5, 4/5)]
- What do you think might be missing from this in-text citation?
- Here is a resource if you'd like to review the full rule. [If students repeated the question or had uncertainty, *Thamus* will provide a watermarked example showing the correct format for reference]
- Would you like to revisit any other areas before submitting?

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